Name: Trịnh Gia Hiệp - 2352343 - HW1

1. ML Problem Specification for Automated Envelope Routing:

a/ Input: An image at the front of the envelope that has an address (either handwritten or printing).

b/ Target: Extract the destination text (ex: ZIP code)

c/ Decision made from the prediction: Sort and transfer that envelope to the corresponding container of that area (region)

d/ Primary Metric: Accuracy because the more accurate of the prediction, the better of the productivity. It takes time to check again and transfer back if the envelope is sorted in a wrong place.

e/ Label noise: When writing the wrong address or bad handwritting or maybe because of typing error.

- The same raw data can support either classification task or regression task based on the decision objective:

Decision Objective:
- Classification: If the postal service wants to separate the envelopes into a discrete number of containers or bins, the model will predict which label
- Regression: If they want to work with a continuous number (like ZIP code or address number)

2. 120 labelled data with 2 plans:

(i) One 80/20 split: Train on 96 examples and test on 24 examples.

- What it tells: The data for training is not many therefore it's hard for the model to generalize good enough, also take 80% of the data in one-shot may miss some good training example from the test set (instead of 5-fold).
- What it does not tell: The model has high variance because of the limited test set, in this case 1 wrong prediction decreases ≈ 4,17%. The model is not stable.

(ii) 5 fold Cross-validation: Split the data in 5 blocks, use 1 block (24 examples) for testing and the rest 4 blocks (96 examples) for training and repeat 5 times total.

- What it tells: For a limited dataset like this, K-fold (5-fold in this case) works best to choose the best model possible, reliable for model selection.
- What it does not tell: Eventhough it's better than 80/20 split but the model still has high variance because of the size of the training and test set.

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Predicted / Actual	Positive	Negative
Positive	36	~~30~~
Negative	14	170

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{36+170}{250} = 0,824$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{36}{36+30} = \frac{6}{11} \approx 0,546$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{36}{36+14} = \frac{18}{25} = 0,72$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{\frac{6}{11} \cdot 0,72}{\frac{6}{11} + 0,72} \approx 0,621$$

- Accuracy may be misleading because the dataset is imbalance, because the condition is very rare, the model can just guess all negative and still has a good accuracy.
- Error type deserved attention: False negative or Recall because it's better to predict false positive to the patient than predicted false negative, if we predicted false negative it's very costly compared to false positive.

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4. Model A (24%, ~~26~~%):
→ Undersitting (high bias): The model is too simple so it cannot capture the pattern in the data
→ Intervention: Increase model complexity.

Model B (3%, 21%)
→ Overfitting (high variance): The model is too complex and memorized most of the training data, therefore training error is low and testing error is high.
→ Intervention: Add Regularization or increase training data.

Model C (9%, 11%)
→ Balance, good generalization.

- As model complexity increases, bias decreases and variance increases:
  - Bias decrease: The model learns the pattern better, compared to simple model that cannot learn the pattern (high bias)
  - Variance increases: Complex models are sensitive to minor fluctuations or maybe errors, compared to simple models may remain more stable.

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5. Protocol card: UCI HAR Using Smartphones (dataset id 240)

- Usecase: To predict ~~physical~~ current physical activity of a person using smartphone inertial sensor signals

- frozen target: Activity label with 6 classes: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING. One label per 2,56s window.
- Split policy: Dataset's own subject-wise split: 21 subjects for train, 9 subjects for test set.
- Primary metric: Macro-F1 so that hard classes count as much as easy ones. The baseline predict the majority class. ~~H~~ (laying).
- Leakage risk: ~~Neighbor~~ Neighbouring overlap by 50% and come from the same person
  → So split by subject, fit scaling and feature selection on training folds.
- Fairness rule: All model has the same folds, same features, seed 42.
- Limitation: Only 30 volunteers with one phone model
  → Model may not generalize to many people, devices.

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